Math 431 – Spring 2026

Homework 12 – Solutions

9.2 (a) We have $E[X] = \frac{1}{\lambda} = 2$ and $X \geq 0$. By Markov's inequality

$$P(X > 6) \leq \frac{E[X]}{6} = \frac{1}{3}.$$

(b) We have $E[X] = \frac{1}{\lambda} = 2$, $\mathrm{Var}[X] = \frac{1}{\lambda^2} = 4$. By Chebyshev's inequality

$$P(X > 6) = P(X - E[X] > 4) \leq P(|X - E[X]| > 4) \leq \frac{\mathrm{Var}(X)}{4^2} = \frac{4}{4^2} = \frac{1}{4}.$$

(c) The exact probability is

$$P(X > 6) = \int_6^\infty \lambda e^{-\lambda x} dx = e^{-6\lambda} = e^{-3} \approx 0.0498.$$

9.6 Let X_k denote the time needed in seconds to it the kth hot dog, and denote by S_n the sum $X_1 + \cdots + X_n$. Since 15 minutes is 900 seconds, we need to estimate the probability $P(S_{64} < 900)$. By the CLT the standardized random variable $\frac{S_{64} - 64 \cdot 15}{\sqrt{64 \cdot 4^2}}$ is close to a standard normal. Thus

$$\begin{aligned} P(S_{64} < 900) &= P\left(\frac{S_{64} - 64 \cdot 15}{\sqrt{64 \cdot 5^2}} < \frac{900 - 64 \cdot 15}{\sqrt{64 \cdot 4^2}}\right) \\ &\approx \Phi\left(\frac{900 - 64 \cdot 15}{\sqrt{64 \cdot 4^2}}\right) = \Phi(-1.875) = 1 - \Phi(1.875) \\ &\approx 0.0304, \end{aligned}$$

where we used linear interpolation to approximate $\Phi(1.875)$ using the table in the Appendix. (Using the value for 1.87 or 1.88 is also acceptable.)

9.18 (a) The random variable X is nonnegative, and from Example 8.13 we have $E[X] = 100 \cdot \frac{1}{\frac{1}{3}} = 300$. Hence by Markov's inequality we get

$$P(X > 500) \leq \frac{E[X]}{500} = \frac{300}{500} = \frac{3}{5}.$$

(b) Again, from Example 8.13 we have $\mathrm{Var}[X] = 100 \cdot \frac{1 - \frac{1}{3}}{(\frac{1}{3})^2} = 600$. Then from Chebyshev's inequality:

$$\begin{aligned} P(X > 500) &= P(X - E[X] > 500 - 300) \\ &\leq \frac{\mathrm{Var}(X)}{200^2} = \frac{600}{200^2} = \frac{3}{200} = 0.015. \end{aligned}$$

(c) We can represent the negative binomial as the sum of iid geometric random variables (see Example 8.13 for more detail). By the CLT the distribution of the standardized version of X is close to that of a standard normal. The standardized version is $\frac{X - 300}{\sqrt{600}}$, hence

$$P(X > 500) = P\left(\frac{X - 300}{\sqrt{600}} > \frac{500 - 300}{\sqrt{600}}\right) \approx 1 - \Phi(\tfrac{20}{\sqrt{6}}) \approx 1 - \Phi(8.16) < 0.0002.$$

Note that 8.16 is not in our table of values for Φ, so the best we can say is that $1-\Phi(8.16) < 1-\Phi(3.49) \approx 0.0002$. In fact $1-\Phi(8.16)$ is way smaller than 0.0002, it is approximately $2.2 \cdot 10^{-16}$.

(d) We need more than 500 trials for the 100th success exactly if there are at most 99 successes within the first 500 trials. Thus denoting by S the number of successes within the first 500 trials we have $P(X > 500) = P(S \leq 99)$. Since $S \sim \text{Bin}(500, \frac{1}{3})$, we may use normal approximation to get

$$P(S \leq 99) = P\left(\frac{S-\frac{500}{3}}{\sqrt{500 \cdot \frac{2}{9}}} \leq \frac{99-\frac{500}{3}}{\sqrt{500 \cdot \frac{2}{9}}}\right) \approx \Phi\left(\frac{99-\frac{500}{3}}{\sqrt{500 \cdot \frac{2}{9}}}\right) \approx \Phi(-6.42) < 0.002.$$

Again, the real value of $\Phi(-6.42)$ is a lot smaller than 0.0002, it is approximately $6.8 \cdot 10^{-11}$.